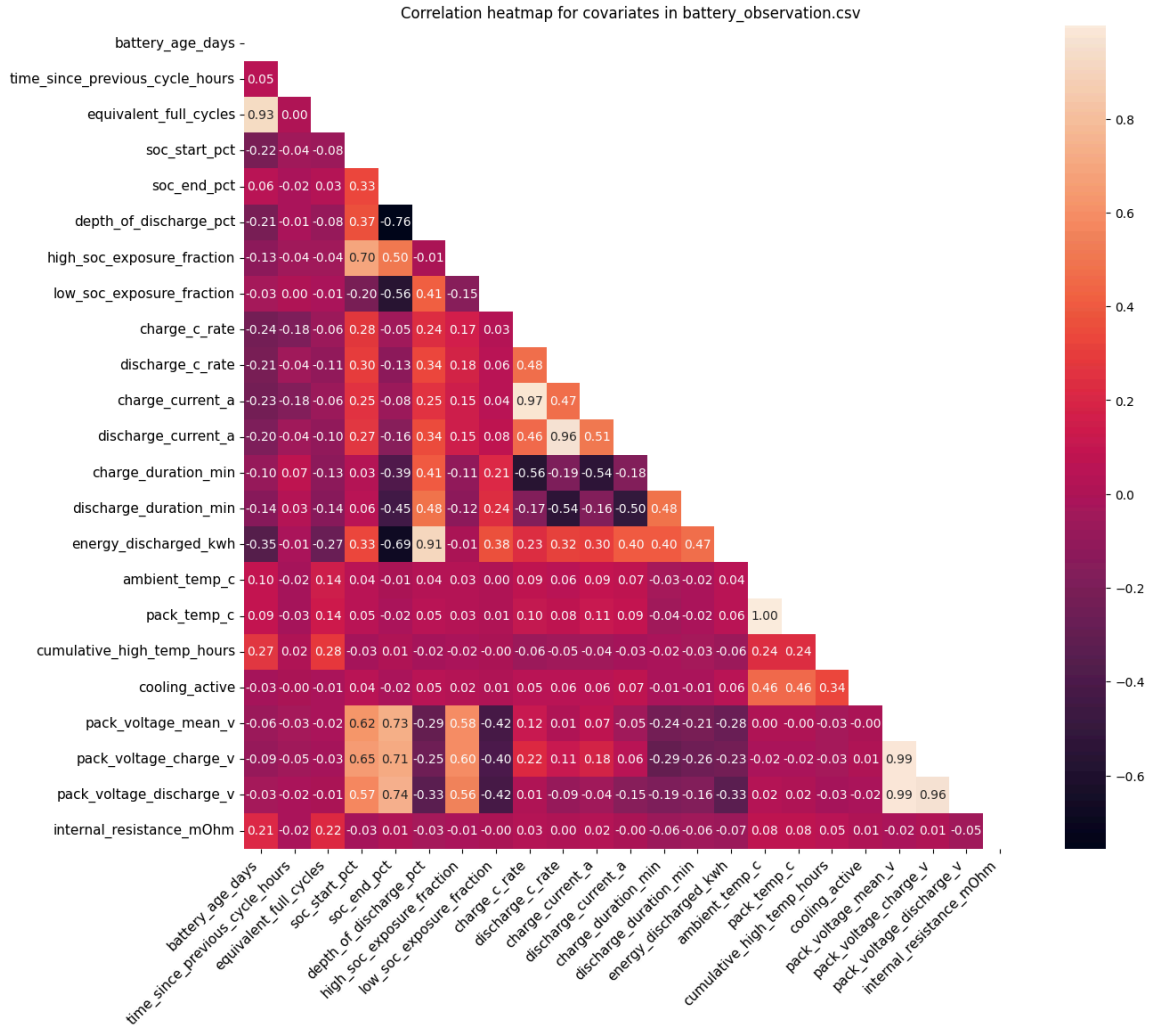
CS342 Coursework

In this coursework the aim is to predict a classification for ‘degredation_status’ which can be (‘Normal’, ‘Moderate’, ‘Severe’, ‘Critical’)

Predicting Battery Useful Life

Task 1: Exploratory Data Analysis and Data Visualisation

Analyse the correlation between features using a heatmap, only using upper triangle since it is symmetric



The correlation heatmap indicates some multicollinearity

The standard practice to deal with multicollinearity is to combine some of the covariates into a single feature for the model.

Task 2: Unsupervised (Classic) Machine Learning for Battery Degradation Analysis

Task 3: Supervised (Classic) Machine Learning for Battery Degradation Analysis

Task 4: Unsupervised Deep Learning for Battery Degradation Analysis

Task 5: Supervised Deep Learning for Battery Degradation Analysis